

Mini Project

**SAS code:
LAB3 (30%)**

Image Recognition and Classification

1. Introduction

Deep learning has revolutionized the fields of computer vision and image processing, enabling highly accurate recognition and classification of complex visual data. This project deepens your understanding of techniques for image recognition and classification, providing hands-on experience in applying these methods to a real-world problem of your choosing.

The lab exercises on AI with Edge Computing — specifically the Rock-Paper-Scissors hand gesture classifier built using InceptionV3 and TensorFlow — serve as your foundation. You will extend this into a fully customized image recognition system.

2. Project Format

This is a group assessment with individual components. Students are required to work in groups with a maximum of 3 members. Collaborative work provides an opportunity to share knowledge, learn from one another, and strengthen teamwork skills, ultimately enhancing the overall quality of the project.

3. Dataset

You are responsible for preparing your own dataset. You may use mobile applications or devices to capture images of real or synthetic objects. It is crucial that your own training and testing datasets are sufficiently large and diverse to avoid underfitting. You must justify the size and balance of the dataset you create.

For reference, the Rock-Paper-Scissors dataset used in the lab contains 2,892 images of diverse hands across different races, ages, and genders, posed into Rock, Paper, or Scissors and labelled accordingly. Use this as a benchmark for the scale and diversity you should aim for.

Your dataset should similarly cover a variety of conditions — different lighting, angles, backgrounds, and subjects — to ensure your model generalizes well to unseen data. When building your dataset, ensure that your images are correctly labelled to facilitate effective training and testing. Proper labelling is essential for guiding the model during the learning process, as it directly impacts the accuracy and performance of the model.

4. Progression

Stage 1 — Replicate and Understand the Lab Exercise

- Start by studying the Artificial Intelligence with Edge Computing Lab Exercises available on BrightSpace. The lab demonstrates a pre-trained InceptionV3 model designed to recognize hand gestures for the Rock-Paper-Scissors game. Watch the demonstration video to understand the full pipeline: dataset preparation, model building, training, saving, and inference.
- The lab pipeline follows these steps: understand and define the problem statement, gather the dataset, create a Python file to load the dataset, build and train the machine learning model, save the model, and predict the result.
- Make sure you can successfully run the Streamlit web app (`rps_app.py`) before proceeding.

Stage 2 — Adapt the Pipeline to Your Own Dataset

- Replace the Rock-Paper-Scissors dataset with your own. Modify the code to reflect your number of classes — for example, if your project involves four categories, change the output layer from `Dense(3, activation='softmax')` to `Dense(4, activation='softmax')` and update the prediction logic accordingly.
- Organise your dataset into the same folder structure used in the lab.
- Use Keras's `ImageDataGenerator` for image augmentation during training. The lab exercise applies a rotation range of 30°, zoom range of 0.2, width and height shift ranges of 0.1, and a validation split of 0.15. Tune these parameters to suit your dataset and document your choices.

Stage 3 — Explore Alternative Pre-Trained Models

- Inception v3 is a widely-used image recognition model that has been shown to attain greater than 78.1% accuracy on the ImageNet dataset. You are encouraged to explore replacing InceptionV3 with other commonly used pre-trained models such as MobileNetV2, ResNet50, or EfficientNet. Compare the training time, accuracy, and overall performance of the models. Present your findings and conclusions.

Stage 4 — Build a Custom CNN from Scratch

- A key part of the project involves building a customized Convolutional Neural Network (CNN) model from scratch. The hidden layers of a CNN typically consist of convolutional layers, pooling layers, fully connected layers, and normalization layers. Construct these layers step by step using TensorFlow or PyTorch. You are free to refer to the Computer Vision Lab Guides or other online resources, with proper citations.
- Your custom CNN should be compared against the transfer learning approach in terms of accuracy, training time, and generalization.

5. Setup Requirements

Download and install Anaconda (Individual Edition), the 64-bit graphical installer with Python 3.x. Open Anaconda Prompt as Administrator and install the following packages:

- Streamlit: `pip install streamlit` (estimated 1 minute)
- TensorFlow 2.0 and above: `pip install tensorflow` (estimated 5–10 minutes)

6. Code Quality

Your code should be well-organized, clear, and properly commented on where necessary to enhance readability and maintainability. While consulting online resources is encouraged, it is essential to provide proper citations for any code or solutions you referred to, including any use of AI tools as per the Academic Integrity declaration.

7. Video Presentation

You are required to prepare a video presentation using MS Teams, with each student presenting for 5 minutes. In your presentation, you should:

- Explain the dataset preprocessing methods and exploratory data analysis techniques you applied, along with the rationale for your choices.
- Walk through the process of building your model, describing the module construction and any modifications made.
- Discuss and analyze the performance of your model, including key performance indicators such as accuracy and loss. Include details on techniques used to improve model performance and prevent overfitting.
- Divide tasks collaboratively: dataset preparation, model building, graph visualization, hyperparameter tuning, optimizer selection, conclusions, and recommendations.

A Q&A session will follow your presentation to assess your understanding and depth of learning. All team members must have a complete working program on their laptops, including the training and inference code as well as the full dataset. Failure to do so will result in penalties for the team.

8. Submission of Work

Due Date: Please upload your work by midnight on Sunday, **2nd August 2026**. No extensions will be granted. A late submission will incur a 10% deduction in total marks per day. Any incomplete submission will also result in a deduction of marks.

Submit your work to the respective submission links. Your submission should include:

(1) Presentation Video

Name your video file as: `1stStudentName_2ndStudentName_3rdStudentName.mp4`

(2) A Zip file containing:

- Signed AI (Academic Integrity) Declaration Form
- Presentation slides
- Training and inference program (.ipynb)
- Requirements (e.g., library dependencies, installation requirements, Python version)

Name the zip file as:

`Program_1stStudentName_2ndStudentName_3rdStudentName.zip`

(3) Dataset

Compress your dataset files and name the zip file as:

`Dataset_1stStudentName_2ndStudentName_3rdStudentName.zip`

Note: BrightSpace only allows files under 2GB to be uploaded. If your dataset exceeds this size, please split among your group members.

9. Marking Scheme

Assessment Components	Marks Allocated
Datasets Preparation (using phone/camera to capture images)	10
Dataset Expansion (splitting data into Training, Validation, and Test sets)	10
Model Building, Training, and Comparisons	25
Data Visualization (using Matplotlib for graphs such as loss vs. epochs, accuracy vs. epochs)	10
Real-Time Image Recognition (live testing on the spot)	10
Video Presentation (10-15 minutes per group)	15
Reflection and Q&A	5
Submission of Work	5
Extra "WOW" Factor (e.g., innovative approach or additional features)	10
Grand Total	100

Declaration of Academic Integrity

Cheating and plagiarism are serious offences at Singapore Polytechnic and may result in expulsion. AI tools may be used provided proper acknowledgement is given and outputs are not replicated wholesale – failure to disclose AI use will be treated as academic dishonesty.

Please complete and sign the Academic Integrity Declaration Form included in your submission.

Module Code & Module Name	
Assessment (e.g. Mini Project)	

Declaration of Academic Integrity

Academic Integrity is a central tenet of Singapore Polytechnic. The polytechnic rules state that *"Cheating in examinations and other assessed work is a very serious offence. This includes copying and using plagiarised material. Any student who cheats, attempts to cheat or breaches any rules for examinations and tests will face disciplinary action. The student is liable to be expelled."*

Check only one of the two options below:

☐ I/We affirm that the work I/we submit is my/our own, produced without help from any AI tool(s).

☐ I/We affirm that the work I/we submit has been produced with the use of AI tool(s) which I/we have acknowledged fully.

By signing this form, I/we declare that the above affirmation made is true, and that I/we have read and understood the rules stated in Students Handbook on “[Plagiarism](#)” and “[Breach of Examination/Assessment Rules](#)”.

#	Student ID	Name	Signature	Date
1				
2				
3				
4				
5				

Important:

Students will not incur penalties for the utilisation of AI tools in their assignments, provided that due acknowledgment is given, and wholesale replication of the AI tools' output is abstained from. Instances where students incorporate AI tools in their assignments and replicate the generated output wholesale, or fail to comprehensively disclose the use of AI tools, will be deemed as instances of academic dishonesty and will be subject to appropriate disciplinary measures.

Acknowledgement

To acknowledge the use of an AI tool, use the template below. Fill in the name of the AI tool, your name, your input prompts, date, and output generated by the AI tool. Extend the template if needed.

Name of AI tool	< For example, ChatGPT >
Name of student	< Insert the name of student who used the AI tool >
Input prompt	< Insert the question that you asked ChatGPT >
Date generated	< Insert the date that ChatGPT response was generated, since ChatGPT is an evolving technology >
Output generated	< Insert the response verbatim from ChatGPT >
Impact on submission	< Briefly explain which part of your submitted work was ChatGPT's response applied >

Name of AI tool	< For example, ChatGPT >
Name of student	< Insert the name of student who used the AI tool >
Input prompt	< Insert the question that you asked ChatGPT >
Date generated	< Insert the date that ChatGPT response was generated, since ChatGPT is an evolving technology >
Output generated	< Insert the response verbatim from ChatGPT >
Impact on submission	< Briefly explain which part of your submitted work was ChatGPT's response applied >

Name of AI tool	< For example, ChatGPT >
Name of student	< Insert the name of student who used the AI tool >
Input prompt	< Insert the question that you asked ChatGPT >
Date generated	< Insert the date that ChatGPT response was generated, since ChatGPT is an evolving technology >
Output generated	< Insert the response verbatim from ChatGPT >
Impact on submission	< Briefly explain which part of your submitted work was ChatGPT's response applied >